

Course Code: CSE 499A

Section: 15

Project Group No: 06

Group Members:

Md.Murad Ul Momin (ID: 2232360642)

Rima Saha (ID: 2132517042)

Sakia Tabassum Ananna (ID: 2221433042)

Ashraful Hossain Riyan (ID: 2232845642)

PRIVATE SHIELD: Weekly Report

I. INTRODUCTION

This week, the team implemented three federated-learning research approaches for the PRIVATE SHIELD IoT intrusion-detection project. The work focused on dynamic aggregation for heterogeneous attack data, defence against free-rider clients, and semi-supervised anomaly detection. Each paper was developed as a separate experimental module so that its core method could be executed and later compared in a common project environment.

II. IMPLEMENTATION WORK

A. FedMADE Dynamic Aggregation

The FedMADE implementation was prepared for federated intrusion detection when IoT clients contain different traffic and attack-class distributions. Local client models were trained independently, and a Class Probability Matrix was produced from validation predictions. Similar models were grouped using DBSCAN. Dynamic aggregation weights were then calculated so that models with stronger class-level contributions received greater influence in the global model. A basic suspicious-client test was also added so that poisoned or poorly performing updates could receive a very small aggregation weight.

B. WEF-Defense Against Free Riders

The WEF-Defense implementation focused on clients that try to receive the global model without performing genuine local training. Changes in penultimate-layer weights were monitored during local epochs, and a dynamic threshold was used to count meaningful changes. These counts formed a Weight Evolving Frequency Matrix for each client. The server compared the matrices using Euclidean distance, cosine similarity, and average frequency. Normal contributors and suspected free riders were then separated, and different aggregation outputs were prepared for the two groups.

C. FedMSE Semi-Supervised Detection

The FedMSE implementation used N-BaIoT-compatible data and normal-only training. The core SAE-CEN structure combined a Shrink Autoencoder with a centroid-based one-class classifier. The autoencoder learned a compact representation of normal traffic, while centroid distance was used to identify abnormal traffic. The MSEAvg aggregation method assigned greater importance to local models with lower reconstruction error. Client datasets were divided into non-IID gateway partitions to represent different IoT devices and traffic conditions.

III. INTEGRATION AND CURRENT STATUS

The three implementations now provide working baselines for robust aggregation, free-rider defence, and anomaly-based intrusion detection. The core training flow, client-server communication logic, and evaluation steps were checked through basic local runs. Exact research-paper performance values are not claimed because the experiments use selected dataset portions and the team's available computing environment.

FUTURE OBJECTIVES

- Run all three methods with the same client count, data split, training rounds, and random seed.
- Compare accuracy, precision, recall, F1-score, AUC, communication time, and attack robustness.
- Test FedMADE under model poisoning and WEF-Defense under multiple free-rider ratios.
- Integrate the strongest components into the main PRIVATE SHIELD detection pipeline.

REFERENCES

- [1] S. Sun, P. Sharma, K. Nwodo, A. Stavrou, and H. Wang, "FedMADE: Robust Federated Learning for Intrusion Detection in IoT Networks Using a Dynamic Aggregation Method," arXiv:2408.07152, 2024.
- [2] J. Chen, M. Li, T. Liu, H. Zheng, Y. Cheng, and C. Lin, "Rethinking the Defense Against Free-rider Attack From the Perspective of Model Weight Evolving Frequency," arXiv:2206.05406, 2022.
- [3] V. T. Nguyen and R. Beuran, "FedMSE: Semi-supervised Federated Learning Approach for IoT Network Intrusion Detection," Computers & Security, vol. 151, Art. no. 104337, 2025.